Excel OCR Cataloger

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Disclaimers:

1) This tool utilizes open-source libraries to function. These open-source libraries are:

- sys - Python Standard Library
- os - Python Standard Library
- pytesseract -
- subprocess - Python Standard Library
- PIL -
- enum - Python Standard Library
- shutil - Python Standard Library
- openpyxl -
- functools - Python Standard Library
- PyQt6 -

The licensing for these libraries can be found at the end of this document under “Credits”.

2) This tool uses OCR to determine the content of the image, however, while OCR can be fairly accurate, it is not perfect. The consistency and quality of the output document rely heavily on the quality of the image and tolerance of the OCR.

3) This tool will install Google’s Tesseract OCR model on the system if it is not already present.

4) The image format should be .JPG or .PNG.

1. Software Setup:

Ensure Python is installed on the device.

If Google Tesseract is not installed, run the program as administrator. The program will automatically install Google Tesseract.

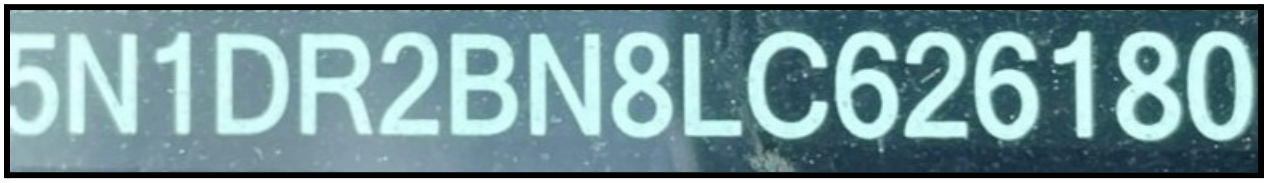
2. Taking & Preparing Pictures:

All pictures should be input in ".PNG" or ".JPG" to avoid errors.
The images containing the serial number must be taken in a particular manner.
Only the serial number should be shown. Crop everything else out.

Bad serial image example:

- Too much information is shown
- Serial value is obscured by glare.

Good serial image example:



- Only the serial value is shown.
- Values are unobscured.
- The border is removed.
- The title is removed.

3. Input Folder Format:

The folder containing all the images for processing can be formatted in two different ways.

Method 1: Format the folder in the order:

"1_model.png, 2_serial.png, 3_model.png, 4_serial.png, …"

Where "(n)_model" is the image of the unit and "(n+1)_serial" is the image of the serial.

Method 2: Format the folder in the order:

"1_serial.png, 2_model.png, 3_serial.png, 4_model.png, …"

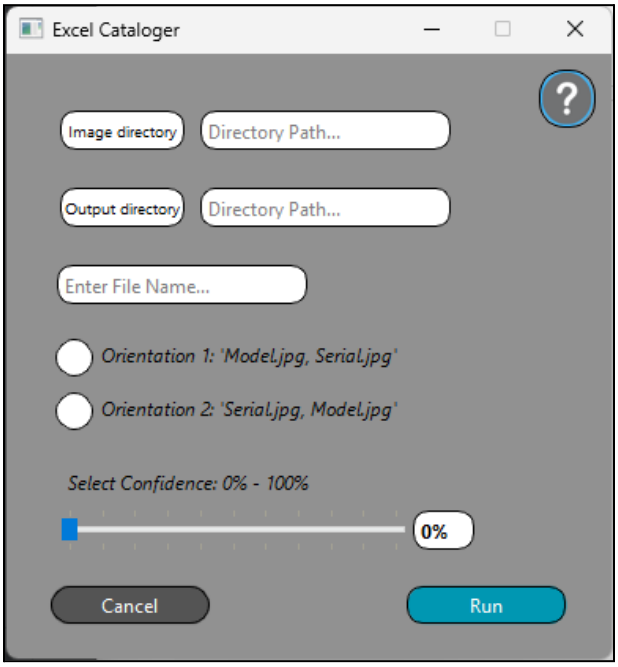
Where "(n)_serial" is the image of the serial and "(n+1)_model" is the image of the unit.

Example format for method 1:

Name	Date	Type	Size
1 model	3/17/2022 5:40 AM	JPG File	130 KB
2 serial	3/11/2024 9:05 PM	PNG File	6 KB
3 model	3/17/2022 5:40 AM	JPG File	130 KB
4 serial	3/11/2024 9:04 PM	PNG File	8 KB
5 model	3/17/2022 5:40 AM	JPG File	130 KB
6 serial	3/11/2024 9:44 PM	PNG File	45 KB
7 model	3/17/2022 5:40 AM	JPG File	130 KB
8 serial	3/11/2024 9:06 PM	PNG File	32 KB
9 model	3/17/2022 5:40 AM	JPG File	130 KB
10 serial	3/11/2024 9:06 PM	PNG File	31 KB
11 model	3/17/2022 5:40 AM	JPG File	130 KB
12 serial	3/11/2024 9:16 PM	PNG File	119 KB

4. Configuring & Running The Program:

If the program is correctly installed and running it should open a window looking like this:



Each function operates with the assumption the formatting of the input folder is correct.

Image Directory:The location of the folder containing the input images

Output Directory: The location the program creates a new folder.

File Name: The name of the folder you want to create.

Orientation Selection: Image format orientation selection (refer to section 3).

Confidence: The confidence threshold you want the program to operate at (refer to section 5).

5. Errors & Tolerance:

Errors:

If the program is not confident in the value returned, it will instead return an error value to the Excel sheet. The program will also return the assumed value appended to the error log. If too many errors occur when compiling a folder of images try either reformatting the images to match the example in section 2 or reducing the confidence level to allow fewer restrictions.

Tolerance:

Because this program uses a machine learning model, errors may occur. To reduce the amount of false positives or false negatives, the user may reduce or increase the confidence level the model operates at. A higher number threshold results in the program only accepting values with a higher confidence of correctness. On the contrary, a lower threshold results in the program accepting values with a lower confidence of correctness.

Summarized:

- Higher values result in more false negatives.
- Lower values result in more false positives.

Credits & Licensing:

Python Standard Library:

© Copyright 2001-2024, Python Software Foundation.
License: https://docs.python.org/3.10/license.html#psf-license

Pytesseract:

© Copyright 2001-2024, Python Software Foundation
License: http://www.apache.org/licenses/LICENSE-2.0
Project Source: https://pypi.org/project/pytesseract/

Google Tesseract:

© Copyright 2006-2024, Google Inc.
License: http://www.apache.org/licenses/LICENSE-2.0
Google's Tesseract-OCR Engine: https://github.com/tesseract-ocr/tesseract

Leptonic Library:

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License: https://creativecommons.org/licenses/by/3.0/us/
Project Source: https://github.com/DanBloomberg/leptonica

PIL:

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License: https://github.com/python-pillow/Pillow/blob/main/LICENSE
Project Source: https://github.com/python-pillow/Pillow/tree/main

Openpyxl:

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Project Source: https://github.com/fluidware/openpyxl/tree/master

PyQt6:

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